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# Premises for the theory of forgotten effects

**Salvador Linares-Mustarós[1*], Anna M. Gil-Lafuente[2], Dolors Corominas Coll[1], Joan C. Ferrer-Comalat[1]**

[1] Department of Business Administration, University of Girona, Carrer de la Universitat de Girona, 10, 17071 Girona, Spain
{salvador.linares; dolors.corominas; joancarles.ferrer}@udg.edu

[2] Department of Business Administration, University of Barcelona, Av. Diagonal 690, 08034 Barcelona, Spain
amgil@ub.edu

*Corresponding author

## Abstract

This paper aims to present the logical foundations in research into forgotten effects by showing an essential corollary to understanding forgotten effects models. The corollary states that any influence that may occur between two events is really of a value greater than or equal to the maximum truth value of any chain of cause and effect between the two events. This is consistent with the fact that in practical research on forgotten effects, experts always change the value of an impact if they are shown a causal chain of more value than they have estimated directly.

Keywords: forgotten effects, computer software, incidence matrices.

## 1. Introduction

The idea of using fuzzy incidence composition to detect the possible non-obvious influences of a cause on an effect is known in the economic literature as "Forgotten Effects Research". The application of incidence matrices in an economic context was first presented to the scientific community in 1988 by Professors Arnold Kaufmann and Jaime Gil Aluja in the book published by Milladoiro "Models to research forgotten effects" (Kaufmann and Gil Aluja, 1988).

There are a great number of academic papers (García, 2001; Mallo et al., 2005; Domech et al., 2008; Gil-Lafuente et al., 2009; Rico, 2010; Gil-Lafuente & Bassa,

2011; Gil-Lafuente et al., 2012; Linares et al., 2013) and research projects (Lladó et. Al., 2000; Gil-Lafuente, 2011, 2012) addressing indirect incidence values based on various models presented by researchers.

Our aim in this paper consists in presenting the logical foundations in research into forgotten effects.To achieve our objective, the chapter has been divided into two sections. The first section presents the terminology used throughout this work and an overview of how the method usually functions. The second section presents the basis of binary logic that supports the theory of research into forgotten effects. The paper ends with conclusions and the bibliography.

## 2. Presentation of the usual method employed to search for forgotten effects based on estimates provided by experts

Analysis of the concept of fuzzy relationships was first undertaken by Zadeh in the 1971 article "Similarity relations and fuzzy orderings" (Zadeh, 1971), extending ideas conceived a few years previously in the seminal article "Fuzzy logic" (Zadeh, 1965).

In 1988, Arnold Kaufmann and Jaime Gil Aluja established (Kaufmann and Gil, 1988) the "Theory of Forgotten Effects" using models which allow all incidence relationships or direct and indirect causalities to be obtained without the possibility of omitting elements that might initially have been fully or partially overlooked.

The following summarizes the general fuzzy rectangular matrix model that will be used to model the specific problem in the following section.

Let A be a set of elements, A = {ai / i=1, 2,..., n}, which we shall call causes, and B, a second set of elements, B = {bj / j=1, 2,..., m}, which could possibly be the same set as A, and we shall call effects.

The resulting set of pairs of elements denoted by the expression v(ai ,bj) shall define what tends to be known as the "direct incidence matrix". This matrix shows the cause-effect relationships that occur between the elements of set A (causes) and those in set B (effects). Said matrix will be denoted by [ $\underline{M}$ ].

Now let C be a third set of elements, C = {$c_k$ / k=1, 2,..., p}, which act as effects of set B (their acting as causes of set A would be treated similarly).

Let [ $\underline{N}$ ] be the incidence matrix that shows the cause-effect relationships occurring between the elements of set B and those of set C.

The mathematical operator that establishes the incidences of A on C is the max-min composition of matrices, whose causal relationship of $a_i$ over $c_k$ is:

$$v(a_i, c_k) = \max_{b_j}\left(\min\left(v(a_i, b_j), v(b_j, c_k)\right)\right)$$

Causes-causes will be noted in the incidence matrix for the whole as [ $\underline{A}$ ] and effects-effects as [ $\underline{B}$ ]. The composition [ $\underline{M}$ *] =[ $\underline{A}$ ]o [ $\underline{M}$ ]o[ $\underline{B}$ ] between these three types of matrices is designed to obtain a new matrix of incidences from among the elements of set A (causes) and those of set B (effects) to reflect any possible indirect causal relationships that may not be taken into account in strategic decision processes.

To find possible effects which may have been overlooked, the algorithm proposed by Kaufmann and Gil suggests studying the values of the matrix [ $\underline{M}$ *] - [ $\underline{M}$ ], as the values obtained thus are maximum values of indirect effects.

The fact that values close to one appear in the matrix [ $\underline{M}$ *] - [ $\underline{M}$ ] indicates that the initial incidents have not been correctly estimated as there are some omissions.

Gento, Lazzari and Machado (Gento et. Al., 2001) pointed out that when creating a direct incidence matrix thanks to the method of recovering forgotten effects, the construction of a new direct incidence matrix results in greater internal coherence with respect to the basis for recovering intermediate effects, obtaining as a result “few forgotten effects”.

Given that in problems involving the selection of certain actions that produce certain benefits for a greater number of actions, overlooked incidents can lead to choosing a less efficient set of actions, the aforementioned fact indicates that research into forgotten effects is an essential method for improving the information the decision-maker has to work with.

In studies carried out in the economic sphere, different values tend to be used for incidences of the same cause of the same effect due to ignorance of actual incidence values. These values are provided by a group of experts and aggregated together using special expert management techniques, in order to obtain a single value for each incidence. One of the most common techniques for adding information together in such works is the expertage technique, first proposed by Kaufmann (Kaufmann, 1987) and developed theoretically by the same author and Professor Gil Aluja (Kaufmann & Gil Aluja, 1993).

In most work on forgotten effects it is also common for experts’ valuation of incidence to be a value belonging to the interval [0,1]. Due to this, researchers tend to provide experts with a linguistic table with a numerical scale similar to Table 1, known as a linguistic endecadary scale. The model of this scale was presented by Kaufmann and Gil Aluja in the book “Introduction to the Theory of Fuzzy Subsets in Business Management” (Kaufmann & Gil Aluja, 1986).

| | |
|---|---|
| null incidence | 0 |
| practically null incidence | 0.1 |
| almost null incidence | 0.2 |
| very weak incidence | 0.3 |
| weak incidence | 0.4 |
| intermediate incidence | 0.5 |
| fair incidence | 0.6 |
| considerable incidence | 0.7 |
| strong incidence | 0.8 |
| very strong incidence | 0.9 |
| absolute incidence | 1 |

**Table 1.** Version of the classic endecadary scale

There is also a second endecadary scale to determine whether the values experts assign to incidence are correct. The main difference between this scale, presented at the 16th INTERNATIONAL SIGEF CONGRESS (Bonet et al., 2010), and Kaufmann and Gil Aluja's classic scale, is how the former determines subjective inputs. This is achieved via a maximum two-stage process using a ternary selection system after eliminating extreme cases. In the first phase of allocation, there is reflection on the degree of truth inherent in the statement, choosing from the possibilities low (L), medium (M) and high (H), once the possibilities of completely false (N for null) and completely true (T for total) have been rejected. If situated within L, M or H, in a second phase, the question is asked whether comparing with the other reference elements believed to be in the same group, the element being evaluated is within the subgroup low (l), medium (m) or high (h).

The final allocation is obtained from the table 2 of values.

| N | L | | | M | | | H | | | T |
|---|---|---|---|---|---|---|---|---|---|---|
| | l | m | h | l | m | h | l | m | h | |
| 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |

**Table 2.** Version of the double ternary scale

Thus, for example, if we aim to associate the possibility of whether incidence that improved water treatment in a certain country might lead to the effect of improving health, we can, in the cases where null (N) and all (T) are rejected, as a first step choose between low (L), medium (M) or high (H). Let us imagine that we have chosen high (H). Then we enter a second phase considering whether this high will be very (h), a little (l) or neither one nor the other (m).

We can establish a bijective application h: S → T between the two scales that retains the following order property:

$$h(u) \leq_T h(v) \text{ if and only if } u \leq_S v$$

It can be concluded that an order isomorphism exists between the two endecadary scales, presented in Table 3.

| (S, ≤S) | | (T, ≤T) |
|---|---|---|
| null incidence | 0 | N |
| practically null incidence | 0.1 | Ll |
| almost null incidence | 0.2 | Lm |
| incidence quite close to null | 0.3 | Lh |
| incidence closer to null than full | 0.4 | Ml |
| neither null nor full incidence | 0.5 | Mm |
| incidence closer to full than null | 0.6 | Mh |
| incidence quite close to full | 0.7 | Hl |
| almost full incidence | 0.8 | Hm |
| practically full incidence | 0.9 | Hh |
| full incidence | 1 | T |

**Table 3.** Order isomorphism

## 3. A proposal for basing the theory on sentential logic

**Definition 1.** A declarative sentence is a sentence in the form of a statement.

Consequently, a sentence is declarative if and only if it only affirms or only denies some objectively verifiable fact. Therefore, in the same context, the truth value of these statements never admits various interpretations since it is well established. The truth value of a declarative sentence "A" is denoted by "v(A)" and this may be "T" or “1” if the declarative sentence is true or "⊥" or “0” if it is false.

For example, the legal definition of entering adulthood usually varies between ages 16–21, depending on the region. If we set a city and a time, "Person X is an adult" is a declarative sentence.

**Note 1.** "It’s raining" or "It's raining outside" are two declarative sentences that have the same meaning. In this case, we talk about declarative sentences or propositions being equivalent. To simplify the calculations, we assume that when a set of declarative sentences is formed, there are no two equivalent statements.

**Definition 2.** A declarative sentence is composed if and only if it can be obtained by connecting one or two declarative sentences with one of the following particles: "not", "and", "or" or "if ... then ...".

**Note 2.** The connector "not" is called negation.

**Note 3.** The connector "and" is called conjunction.

**Note 4.** The connector "or" is called (inclusive) disjunction.

**Definition 3.** The connector "if ... then ..." is called conditional. And in this sense, in the expression "if A, then B", "A" is termed the antecedent and "B" the consequent of the conditional.

**Definition. Classical Propositional Calculus.** To formalize the set of composed declarative sentences, the following assumptions are made.

- All sentences formed from the logical connective "not" applied to a declarative sentence are declarative sentences that have the following truth values:

| A | | not A ($\neg$A) |
|---|---|---|
| 1 | | 0 |
| 0 | | 1 |

This table is read, for example, as “If the declarative sentence "A" is true, then the sentence "not A" is a false declarative sentence”.

- All sentences formed from the logical connectors "... and ..." "... or ..." and "if ... then ..." applied to two declarative sentences are declarative sentences whose truth value is one of the following tables:

| A | B | | A and B ($A \wedge B$) |
|---|---|---|---|
| 1 | 1 | | 1 |
| 1 | 0 | | 0 |
| 0 | 1 | | 0 |
| 0 | 0 | | 0 |

| A | B | | A or B ($A \vee B$) |
|---|---|---|---|
| 1 | 1 | | 1 |
| 1 | 0 | | 1 |
| 0 | 1 | | 1 |
| 0 | 0 | | 0 |

| A | B | | If A, then B ($A \rightarrow B$) |
|---|---|---|---|
| 1 | 1 | | 1 |
| 1 | 0 | | 0 |
| 0 | 1 | | 1 |
| 0 | 0 | | 1 |

**Note 5:** Pla (1991) showed that the structure of declarative sentences with connections "not" and "and" lets build "if ... then ..." connection and other new connections, as "or. .. or ... " and " ... if and only if ... ". For example, the "if ... then ..." connection can be obtained with the following formative construction:

| A | B | | $\neg A \vee B$ |
|---|---|---|---|
| 1 | 1 | | 1 |
| 1 | 0 | | 0 |
| 0 | 1 | | 1 |
| 0 | 0 | | 1 |

**Note 6:** The fact of working with 0 and 1 instead of $\perp$ and T allows the possibility of defining the evaluation of declarative sentences connection in terms of nonlinear mathematical functions. For example, the mathematical function of the $\neg A$ operation can be defined as $v(\neg A) = 1 - v(A)$; $A \wedge B$ can be defined as
$v(A \wedge B) = \text{mín} \{ v(A), v(B) \}$; $A \vee B$ can be defined as
$v (A \vee B) = \max \{ v(A), v(B) \}$; and finally, $A \rightarrow B$ can be defined as
$v(A \rightarrow B) = \max \{ 1- v(A), v(B) \}$.

**Proposition 1.** $\wedge$ is associative.

Proof: It's trivial according to this composition table.

| A | B | C | | $A \wedge B$ | $(A \wedge B) \wedge C$ | $B \wedge C$ | $A \wedge (B \wedge C)$ |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | | 0 | 0 | 0 | 0 |

**Proposition 2:** If A, B and C are three declarative sentences, $(A\rightarrow B)\wedge(B\rightarrow C)$ is a declarative sentence.

Proof:

$(A\rightarrow B)\wedge(B\rightarrow C)$ is a sentence composed of the sentences $A\rightarrow B$ and $B\rightarrow C$.

We know from Classical Propositional Calculus that $A \rightarrow B$ and $B\rightarrow C$ are declarative sentences. Consequently, from Classical Propositional Calculus, their connection with $\wedge$ is also a declarative sentence.

**Note 4:** The next proposition allows the modelling of Hume's idea about the connection between two objects related to a third object, admitted by Hume as a principle of human nature. Specifically, Hume wrote "...we must consider, that two objects are connected together in the imagination, not only when the one is immediately resembling, contiguous to, or the cause of the other, but also when there is interposed betwixt them a third object, which bears to both of them any of these relations" (Hume, 1739) .

**Proposition 3:** If A, B and C are three declarative sentences, then $v[A\rightarrow C] \geq v[\ (A\rightarrow B)\wedge(B\rightarrow C)\ ]$.

Proof: It's trivial according to this composition table.

| A | B | C | | $A\rightarrow B$ | $B\rightarrow C$ | $(A\rightarrow B)\wedge(B\rightarrow C)$ | $A\rightarrow C$ |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | | 1 | 1 | 1 | 1 |

**Proposition 4:** If $A_1$, $A_2$, ... $A_n$ are declarative sentences, then

$(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1}) \wedge (A_{n-1} \rightarrow A_n)$ is a declarative sentence.

Proof: by mathematical induction.

Step 1. Show it is true for n=2.
The second proposition shows it is true.
Step 2. Assume it is true for $n-1 \geq 2$.
Then $(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1})$ is a declarative sentence. $A_{n-1} \rightarrow A_n$ is a declarative sentence according to Classical Propositional Calculus. Therefore, according to Classical Propositional Calculus the union ($\wedge$) between $(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1})$ and $A_{n-1} \rightarrow A_n$ is a declarative sentence. So, $(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1}) \wedge (A_{n-1} \rightarrow A_n)$ is a declarative sentence.

**Proposition 5:** If $A_1$, $A_2$, ... $A_n$ are declarative sentences, then

$$v[A_1 \rightarrow A_n] \geq v[(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1}) \wedge (A_{n-1} \rightarrow A_n)].$$

Proof: by mathematical induction.

Step 1. Show it is true for **n=2.**

The third proposition shows it is true.

Step 2. Assume it is true for $n-1 \geq 2$.

Then $v[A_1 \rightarrow A_{n-1}] \geq v[(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1})] =$
$= \max[v[A_1 \rightarrow A_2], \ldots, v[A_{n-1} \rightarrow A_n]]$,
and $v[A_1 \rightarrow A_n] \geq v[(A_1 \rightarrow A_{n-1}) \wedge (A_{n-1} \rightarrow A_n)] = \max[v[A_1 \rightarrow A_{n-1}], v[A_{n-1} \rightarrow A_n]]$ , so,
$v[A1 \rightarrow An] \geq \max[v[A1 \rightarrow An-1], v[An-1 \rightarrow An]] \geq$
$\geq \max[\max[v[A1 \rightarrow A2], \ldots, v[An-1 \rightarrow An]], v[An-1 \rightarrow An]] =$
$= \max[v[A1 \rightarrow A2], \ldots, v[An-1 \rightarrow An], v[An-1 \rightarrow An]] =$
$= v[(A1 \rightarrow A2) \wedge (A2 \rightarrow A3) \wedge \ldots \wedge (An-2 \rightarrow An-1)]$.

**Corollary:** If $A_1$, $A_2$, $A'_2$, ..., $A_{n-1}$, $A'_{n-1}$, $A_n$ are declarative sentences, then

$$v[A_1 \rightarrow A_n] \geq \max(v[(A_1 \rightarrow A_2) \wedge (A_2 \rightarrow A_3) \wedge \ldots \wedge (A_{n-2} \rightarrow A_{n-1}) \wedge (A_{n-1} \rightarrow A_n)],$$
$$v[(A_1 \rightarrow A'_2) \wedge (A'_2 \rightarrow A'_3) \wedge \ldots \wedge (A'_{n-2} \rightarrow A'_{n-1}) \wedge (A'_{n-1} \rightarrow A_n)]) =$$
$$= \max(\min[v(A_1 \rightarrow A_2), v(A_2 \rightarrow A_3), \ldots, v(A_{n-2} \rightarrow A_{n-1}), v(A_{n-1} \rightarrow A_n)],$$
$$\min[v(A_1 \rightarrow A'_2), v(A'_2 \rightarrow A'_3), \ldots, v(A'_{n-2} \rightarrow A'_{n-1}), v(A'_{n-1} \rightarrow A_n)])$$

## 4. Conclusion

This paper has shown that in binary logic any influence that may occur between two events is a truth value greater or equal to the maximum truth value of any chain of cause and effect between the two given events. This idea is essential to understanding the theory of forgotten effects. To test whether this applies to fuzzy logic is the future goal of this study.

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